oscillation) of P is the same as that of Q's.

14 B Increase in EPE = $\frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2 = \frac{1}{2}(25)\left(0.070^2 - 0.055^2\right) = 0.0234$ J

(The increase in EPE is NOT $\frac{1}{2}k\left(x_f - x_i\right)^2$.)

Energy of oscillation = work done by external force to pull it down
= increase in EPE – loss in GPE
= 0.0234 - $(0.140)(9.81)(0.015)$
= 0.00280 J

Note that this energy of oscillation = maximum kinetic energy of the mass. It can also be calculated from $\frac{1}{2}m\omega^2 x_0^2 = \frac{1}{2}kx_0^2$, since $\omega^2 = \frac{k}{m}$.

$\frac{1}{2}kx_0^2 = \frac{1}{2}(25)(0.015)^2 = 0.00281$ J

15 B Air particle at point X is at a point of compression and hence its displacement is zero.